STAFF REVIEW · H446 003 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q6

Do Now — source answers

1. Boolean: True or False.

2. 64 + 16 + 8 + 2 = 90 .

3. +18 = 00010010 . Sign magnitude: 10010010 . Two’s complement: invert → 11101101 , add 1 → 11101110 .

4. 10010000 . 112 + 32 = 144, outside −128…127. Signed overflow occurs; the stored pattern represents −112 when interpreted as an 8-bit two’s-complement integer.

I Do — source worked steps

1. Find the hex digits — CODE:
173 ÷ 16 = 10 remainder 13 — The quotient gives the first hex digit; the remainder gives the second.

2. Convert 10 and 13 to hex — CODE:
10 = A
13 = D

$173_{10} = AD_{16}$

3. Use four bits per hex digit — CODE:
A → 1010
D → 1101

$AD_{16} = 10101101_{2}$

4. Return to denary — CODE:
(10 × 16) + 13 = 173

128 + 32 + 8 + 4 + 1 = 173 — For binary → hex, group bits from the right in fours and pad the left group with zeros if needed.

We Do — guided CHECK

1. Build the answer
Reverse check: 755 ÷ 16 gives remainders 3, F, 2.

2. Build the answer
The third shift discards the leftmost 1, so 352 cannot fit in 8 bits.

3. Build the answer
The mask keeps the low nibble and clears the high nibble.

4. Build the answer

5. Build the answer
44 ÷ 4 = 11; the discarded bits are both zero.

Page 3 — independent skill check, teacher-review model

1. Logical >>2 gives 00101001 (discard two rightmost bits, fill with zeros). AND 00001111 gives 00000101 (clear upper four bits).

You Do — source mark schemes

1. B = 1011; 7 = 0111; E = 1110. Answer: 1011 0111 1110 (1). Retain four bits per hexadecimal digit.

2. 188 ÷ 16 = 11 remainder 12 → BC (1). Check: 11 × 16 + 12 = 188.

3. 1110 = E and 0101 = 5 (1); E5 (1).
(3 × 256) + (10 × 16) + 7 (1) = 935 (1).
214 ÷ 16 = 13 remainder 6 (1); D6 (1).

4. 00001101 (1), 13 (1).
01000010 (1). The leftmost 1 is shifted out and discarded (1); 161 × 2 = 322 exceeds 255, while the stored value is 66 (1).
00000111 (1). The stored value is 7, not 7.5.

5. CODE:
10100101 10100101 10100101
AND 00001111 OR 01000010 XOR 11110000
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00000101 11100111 01010101
One mark per correct result. Each operation starts from the original byte.

6. AND (1) with 00111111 (1). Zeros clear bits 7 and 6; ones preserve the rest.
OR (1) with 00001001 (1). Ones set bits 3 and 0.
XOR (1) with 00100100 (1). Ones toggle bits 5 and 2.
Q1–Q2 retain the original exam mark allocations. Q3–Q6 use classroom allocations. Answers are calculated teaching guidance, not a reproduced official OCR mark scheme.

Exit Ticket — source answer

1. True. Each hexadecimal digit represents four binary bits.